

A context-conditioned loading potential with force obtained by differentiation

a Inputs & conditioning

Material & loading context
(g, T, v)

Grain size · temperature · velocity

Training-set standardization → x

MLP 3 → 128 → 128 → 12

Constrained parameters

k, F_y, a, b, c > 0

δ_h > δ_y > 0; 1 < m < 4

w₁, w₂, b₁, b₂ > 0

Softplus / bounded sigmoid transforms

Indentation depth δ

One shared differentiation variable

b Energy construction

Analytical energy branches

Elastic U_e = (2/5) k d^{5/2}

Plastic U_p = F_yδ + (a/2) Δ² + (b/3) Δ³

Hardening U_h = F_yδ + c Δ^{m+1}/(m + 1)

Δ = softplus₁₀(δ - δ_y) + ε

d = max(δ, ε); ε = 10⁻⁶

Depth-dependent sigmoid gates

π₁ = sigmoid[(δ - δ_y)/w₁ - b₁]

π₂ = sigmoid[(δ - δ_h)/w₂ - b₂]

Learned residual energy U_{res}(δ, x)

MLP [d, d^{3/2}, x] → 128 → 128 → 1

c Force prediction

Gated loading potential

U = Σ_j w_jU_j + U_{res}

w_e = 1 - π₁

w_p = π₁(1 - π₂)

w_h = π₁π₂

Σ_j w_j = 1

Full autograd

$\hat{F} = \partial U / \partial \delta$

Differentiate branches
and residual energy

Predicted force–depth response

Full derivative

j ∈ {e, p, h}

$\hat{F} = \sum_j w_j U'_j$

Branch derivatives

+ Σ_j w'_jU_j

Gate derivatives

+ U'_{res}

Residual derivative

' = ∂/∂δ at fixed context

01 Train the mean model

Force MSE + physics penalties + residual L² regularization

Hertzian response · monotonicity · force positivity · gate timing/order

SiLU hidden activations in both MLPs; context also enters the residual.

02 Fit uncertainty after freezing the mean

[d, d^{3/2}, x] → SigmaHead → σ > 0

Residual Gaussian NLL + scale regularization

Separate MLP: 5 → 96 → 96 → 1